SAT MATH

Algebra • Practice 1.1A

Practice 1.1A • Linear Equations in One Variable

Name: _____ Date: _____

Directions

These questions follow the format of the SAT Math section. A calculator is allowed on every question (Desmos is built into the digital SAT). Multiple-choice questions have one best answer. For **student-produced responses**, write your answer on the line: a fraction or a decimal is accepted (for example $3/4$ or 0.75); give negative answers with a minus sign; do not use mixed numbers.

Question 1

What value of x is the solution to $7x - 4 = 3x + 20$?

(A) 4

(B) 6

(C) $\dfrac{16}{10}$

(D) 24

Question 2

If $2(x + 5) - 3 = 5x - 8$, what is the value of x?

(A) 5

(B) $\dfrac{5}{3}$

(C) -5

(D) 15

Question 3

In the equation $4(3x - 1) = 12x + c$, c is a constant. If the equation has infinitely many solutions, what is the value of c?

(A) 4

(B) -1

(C) -4

(D) 12

Question 4

If $3x + 7 = 22$, what is the value of $6x - 1$?

(A) 5

(B) 29

(C) 30

(D) 43

Question 5

A gym charges a \$50 joining fee and \$25 per month. Awa has paid \$325 in total. Which equation gives the number of months m she has paid for, and what is m?

(A) $50m + 25 = 325$; $m = 6$
(B) $25m - 50 = 325$; $m = 15$
(C) $50 + 25m = 325$; $m = 13$
(D) $50 + 25m = 325$; $m = 11$

Question 6

What is the solution to $\frac{x}{2} + \frac{x}{5} = 14$?

(A) 20
(B) 10
(C) $\frac{98}{10}$
(D) 70

Question 7

STUDENT-PRODUCED RESPONSE

What is the solution to $0.4(x - 5) = 0.3x + 1.2$?

Your answer: _____

Question 8

STUDENT-PRODUCED RESPONSE

In the equation $5(x - k) = 5x - 35$, k is a constant. If the equation has infinitely many solutions, what is the value of k?

Your answer: _____

Question 9

STUDENT-PRODUCED RESPONSE

What value of x satisfies $\frac{2x + 1}{3} = \frac{x + 4}{2}$?

Your answer: _____

Question 10

Which equation has no solution?

(A) $2(x + 1) = 2x + 2$
(B) $3x + 1 = 2x + 1$
(C) $2(x + 1) = 2x + 3$
(D) $4x = 0$